

Digital Forensics Investigation Report

WormLocker 2.0 Ransomware Analysis

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1 Executive Summary

WormLocker was spotted in the 2020 for the first time, it was spreading through emails since back then and targetting many users and organizations. This ransomware is a .NET executable for windows, during its execution it spreads, takes ownership of the system, encrypts the files and persists in the system and displays a ransom note while locking the screen, in addition if you try to reboot or poweroff your machine it will destroy the windows system completely!

1.1 Incident Overview

1.2 Key Findings

- **Malware Identified:** WormLocker2.0
- **SHA-256:** d4c83205cf6f3098ab6a757312525f4d14a57a819306eeea5c0d022b00b38cf3
- **Primary Impact:** Encrypting the files + Screenlock
- **Persistence Established:** Creating Run keys to auto start in every reboot
- **System Modifications:** File encryption and Registry key modifications
- **Dropped Files:**
 - LogonUIinf.exe
 - Ransom_voice.vbs
 - WormLocker2.0.exe
- **Encryption Status:** Completed, all files were encrypted
- **Ransom Delivery:** Full screen ransome in a GUI form

1.3 Investigation Outcome

2 Tools & Environment

2.1 Memory Forensics Tools

Tool	Details
Framework	Volatility 3
Memory Dump	infected-mem.raw
Dump Size	6.6GB
Capture Method	VBoxManage dumpcore

Table 1: Memory Forensics Environment

Tool	Details
Platform	Autopsy 4.21
Disk Image	malware_snapshot.vmdk
Image Size	20GB
File System	NTFS / exFAT

Table 2: Disk Forensics Environment

2.2 Disk Forensics Tools

2.3 Supporting Tools

- **Hashing:** sha-256sum
- **String Analysis:** Text section in Autopsy
- **Malware Intelligence:** VirusTotal and AnyRun

2.4 Analysis Workstation

- Operating System: Windows 10 x64
- RAM: 8GB
- Storage: 100GB

3 Case Timeline

A summary of important timestamps:

Event	Timestamp
Modified:	2025-11-15 16:32:10 CET
Accessed:	2025-11-15 16:32:10 CET
Created:	2025-11-15 16:32:10 CET
Changed:	2025-11-15 16:32:10 CET

Timestamp (UTC)	Event Description
2025-11-15 14:23:15	Initial Compromise
2025-11-15 14:23:47	WormLocker2.0.exe created
2025-11-15 14:23:52	WormLocker2.0.exe first execution
2025-11-15 14:24:03	takeown command executed
2025-11-15 14:24:05	icacls command executed
2025-11-15 14:24:18	ransom_voice.vbs created
2025-11-15 14:24:20	VBScript execution
2025-11-15 14:24:35	Registry Run key modified
2025-11-15 14:24:37	Startup folder entry created

Timestamp (UTC)	Event Description
2025-11-15 14:25:02	Ransom note created
2025-11-15 14:25:10	Screen lock activated
2025-11-15 16:47:22	Memory snapshot captured
2025-11-15 15:15:38	Disk image acquired

Table 3: Incident Timeline

Key Observations:

- The malware first drops its malicious process, then after a while it encrypts the data and blocks the screen and shows the ransom note. The malware attempts to change in registry run keys to make persistence in the system. A reboot lead to crashing the Windows system completely.

4 Memory Forensics - Volatility 3 Analysis

4.1 Image Information

4.1.1 System Profile Detection

Listing 1: Volatility Image Info Command

```
1 vol -f infected-memory.raw windows.info
```

System Information:

Attribute	Value
Operating System	Windows 10
Kernel Version	10.0.19041
Architecture	x64 bit
Number of Processors	4
System Time	2025-11-15 21:16:40+00:00

Table 4: System Profile from Memory

4.2 Running Processes

Listing 2: Extract Process List

```
1 vol -f infected-memory.raw windows.pslist
2 vol -f infected-memory.raw windows.pstree
```

Suspicious Processes Identified:**Analysis Notes:**

Suspicious items:

- WormLocker2.0.exe

PID	Process	PPID	Threads	Status
528	cmd.exe	7508	1	
3680	conhost.exe	528	5	
4276	WormLocker2.0	7508	15	
5380	wscript.exe	4276	4	
7952	dllhost.exe	820	4	

Table 5: Malicious Process Artifacts

- ransom_voice.vbs (via wscript.exe)
- cmd.exe executing takeown + iccls

We can notice that the pid and the ppid of the process are related, meaning that one process is a child of another. These process are the ones responsible for the launch, persistence and the spread of the malware, as well as the encryption of the files, worm.bin is the main file that launches the rest of the prof wscript process is the one who created the ransom note.

4.3 Process Command Lines

Listing 3: Extract Command Line Arguments

```
1 vol -f infected-memory.raw windows.cmdline
```

Malicious Commands Detected:

PID	Command Line
528	C:\Windows\System32\cmd.exe /k takeown /f C:\Windows\System32
528	iccls C:\Windows\System32 /grant "%username%:F"
5380	wscript.exe ransom_voice.vbs

Table 6: Command Line Artifacts

Commands explanaiton:

Listing 4: Extract Command Line Arguments

```
1 "C:\Windows\System32\cmd.exe" /k takeown /f C:\Windows\System32
  && iccls C:\Windows\System32 /grant "%username%:F"
```

Takeown command: This command is responsible for privilege escalation, it grants the malware with ownership for all the files and for the system and making it able to execute other commands and process easily with admin privileges.

iccls command: This command is responsible for giving the full permission to the malware of read write and execute. These two commands when combined, the malware will be able to fully access and modify the files which in normal cases cannot modify or access, and thus facilitating the encryption and reg modification process.

4.4 Network Connections

Listing 5: Scan Network Activity

```
1 vol -f infected-memory.raw windows.netscan
```

Network Connections Found:

Local Addr	Remote Addr	Port	State	PID	Process
------------	-------------	------	-------	-----	---------

Table 7: Network Connection Analysis

No network connections related to the malware process was found, which means the malware did not establish any connections.

4.5 Registry Keys from Memory

Listing 6: Extract Registry Keys

```
1 vol -f infected-memory.raw windows.registry.hivelist
2 vol -f infected-memory.raw windows.registry.printkey --key "
  Software\Microsoft\Windows\ShellNoRoamMUICache\C:Windows\
  System32" --recurse
```

Registry Artifacts Found:

Listing 7: Extract from REGISTRY USER S-1-5-21-1482476501-1645522239-1417001333-500

```
1 \Software\Microsoft\Windows\ShellNoRoam\MUICache\C:\Windows\
  System32\WormLocker2.0.exe
```

4.6 Malicious Script or EXE in Memory

Listing 8: Dump Process Memory

```
1 vol -f infected-memory.raw windows.procdump -p [PID] --dump-dir
  output/
2 vol -f infected-memory.raw windows.dumpfiles --pid [PID]
```

Dumped Files:

- WormLocker2.0.exe: DLL files (a lot) and some processes like cmd.exe and conhost.exe.

4.7 Malicious Code Detection in Memory

Listing 9: Dump Process Memory

```
1 vol -f infected-memory.raw windows.malfind
```

Key findings:

- WormLocker2.0.exe: was detected as malicious due to these findings in 5 memory regions:
 - PAGE_EXECUTE_READWRITE, meaning the malware will be able to read write and execute codes in the memory.
 - Protection disabled, making it easier for the malware to run.
 - Suspicious hexdump in the memory, related to the malware instructions.
 - The malware created private memory parts and marked them as executable, normal programs rarely do this, so it is a malicious sign.

5 Disk Forensics - Autopsy Analysis

5.1 File System Overview

Partition: NTFS / exFAT (0x07): 2048-1048855551

Users: yamina

5.2 Malicious Files Found

List of Malicious Files:

Filename	Path	Size
wscript.exe	C:\Windows\System32	165 KB
WormLocker2.0.exe	/img_malware_snapshot.vmdk/vol_vol2/Windows/System32/WormLocker2.0.exe	22150 KB
LogonUI.exe (modified)	C:\Windows\System32\LogonUI.exe	—

Table 8: Malicious Files Identified

File Metadata:

WormLocker2.0.exe:

- Creation Time: 2025-11-17 16:32:10 UTC
- Modification Time: 2025-11-17 16:32:10 UTC
- Access Time: 2025-11-17 16:32:35 UTC

wscript.exe:

- Creation Time: 2019-03-19 05:45:02 UTC
- Modification Time: 2025-03-09 11:36:45 UTC
- Access Time: 2025-11-17 16:32:35 UTC

LogonUI.exe:

- Creation Time: 2019-03-19 05:44:00 UTC
- Modification Time: 2025-11-17 16:32:10 UTC
- Access Time: 2025-11-17 16:23:26 UTC

5.3 Persistence Mechanisms

LogonUI File Hijacking:

LogonUI is a Windows system file responsible for displaying the lock screen after rebooting the computer. It runs immediately after the restart, which is why malware often hijacks it for persistence mechanisms. Since this file executes at every reboot, it loads the malicious code repeatedly and displays the ransom note.

The following path was extracted from the text data of the hijacked LogonUI.exe:

C:/Users/Worm/source/repos/wormlogon/wormlogon/obj/Debug/wormlogon.pdb

The PDB (Program Database) path reveals the project folder's path on the developer's machine, where Visual Studio stores temporary compiled files during debugging. This serves as proof that the LogonUI file was modified and replaced with malicious code.

Startup Folders:

5.4 Ransom Notes

Ransom Note Details:

The ransom note was dropped by the process wscript.exe

- Location: C:/Windows/System32/ransom_voice.vbs
- Filename: ransom_voice.vbs

5.5 File Modification Evidence

▼ Name	S	C	O	Modified Time	Change Time	Access Time
worm_tool.sys				2025-11-17 16:32:27 CET	2025-11-17 16:32:27 CET	2025-11-18 16:25:49 CET
winrar-x64-713.exe				2025-11-17 16:32:27 CET	2025-11-17 16:32:27 CET	2025-11-17 16:32:27 CET
password.txt				2025-11-17 16:32:27 CET	2025-11-17 16:32:27 CET	2025-11-18 16:25:49 CET
damn				2025-03-23 22:54:22 CET	2025-03-23 22:54:22 CET	2025-11-18 16:25:49 CET
d4c83205cf6f3098ab6a757312525f4d14a57a8193				2025-11-17 16:32:27 CET	2025-11-17 16:32:27 CET	2025-11-17 16:32:27 CET
d4c83205cf6f3098ab6a757312525f4d14a57a8193				2025-11-17 16:31:26 CET	2025-11-17 16:31:26 CET	2025-11-18 16:25:48 CET
[parent folder]				2025-03-18 03:02:06 CET	2025-03-18 03:02:06 CET	2025-11-17 16:19:45 CET
[current folder]				2025-11-17 16:32:26 CET	2025-11-17 16:32:26 CET	2025-11-18 16:25:48 CET

Hex	Text	Application	File Metadata	OS Account	Data Artifacts	Analysis Results	Context	Annotations	Other Occurrences
Strings									
Extracted Text									
Translation									
Page: 1 of 1 Page Go to Page: Script: Latin - Basic									
.DDB5									

Figure 1: File Modified

▼ Name	S	C	O	Modified Time	Change Time	Access Time	Created Time
notepad.exe				2025-03-23 22:40:47 CET	2025-03-23 22:41:05 CET	2025-11-17 16:19:52 CET	2025-03-23 2
decrypt.py				2025-11-17 16:32:27 CET	2025-11-17 16:32:27 CET	2025-11-17 16:32:27 CET	2025-03-23 2
credit_card_infos.txt				2025-11-17 16:32:27 CET	2025-11-17 16:32:27 CET	2025-11-17 16:32:27 CET	2025-03-23 2
[parent folder]				2025-11-17 16:32:26 CET	2025-11-17 16:32:26 CET	2025-11-18 16:25:48 CET	2025-03-09 0
[current folder]				2025-03-23 22:54:22 CET	2025-03-23 22:54:22 CET	2025-11-18 16:25:49 CET	2025-03-23 2
MALWARE-20250323-215420.raw				2025-03-23 22:54:54 CET	2025-03-23 22:54:54 CET	2025-11-18 16:25:49 CET	2025-03-23 2
Dumplt.exe				2025-11-17 16:32:27 CET	2025-11-17 16:32:27 CET	2025-11-17 16:32:27 CET	2025-03-23 2
Comae-Toolkit-v20230117				2025-03-23 22:49:57 CET	2025-03-23 22:50:04 CET	2025-11-17 16:32:27 CET	2025-03-23 2

Hex	Text	Application	File Metadata	OS Account	Data Artifacts	Analysis Results	Context	Annotations	Other Occurrences
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Strings	Extracted Text	Translation
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Page: 1 of 1	Page	Go to Page:	Script: Latin - Basic
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kEB/>

Figure 2: File 2 modified

5.6 User Data Status (Encrypted or Not)

Encryption Analysis:

Files was encrypted without the modification of the extention (no .encrypted added), We can know from inspecting the content of the files that they are clearly encrypted.

Files tested: passwords.txt, credit_card.info.txt

Open and modified at the time of the execution of the process

Extentions: no extention changes was made.

Entropy test results: 7.33003

5.7 Recovered Files

Deleted Files Recovered:

Could not recover files because they were encrypted and not deteled, so the key is crucial or decrypting the files here.

6 Indicators of Compromise (IoCs)

6.1 File IoCs

6.2 Registry IoCs

Registry Keys Modified:

- /WINDOWS/SYSTEM32 — a run program artifact

Filename	Path
WormLocker2.0.exe	/img_malware_snapshot.vmdk/vol_vol2/Windows/System32/ WormLocker2.0.exe
credit_card_info.txt	/img_malware_snapshot.vmdk/vol_vol2/Users/vboxuser/Desktop/ Sdamn/credit_card_infos.txt
password.txt	/img_malware_snapshot.vmdk/vol_vol2/Users/vboxuser/Desktop/ /password.txt
LogonUI.exe	/img_malware_snapshot.vmdk/vol_vol2/Windows/System32 /LogonUI.exe
wscript.exe	/img_malware_snapshot.vmdk/vol_vol2/Windows/System32 /wscript.exe
ransom_note.vbs	/img_malware_snapshot.vmdk/vol_vol2/Windows/ System32/ransom_voice.vbs

Table 9: File-Based Indicators

```

1 HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Policies\System\
  DisableTaskMgr
2 HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Winlogon
3 HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Run
4 HKCU\SOFTWARE\Microsoft\Windows\CurrentVersion\Run

```

6.3 Command-line IoCs

```

1 takeown /f C:\Windows\System32
2 icacls C:\Windows\System32 /grant "%username%:F"

```

6.4 Behavioral IoCs

- Screen lock behavior
- VBScript execution
- System32 permission hijacking
- Boot and Logon Autostart

7 Conclusion

Summerization of the incident

The wormlocker successfully executed, it locked the system and attempted to encrypt files, some variants of this specific malware DOES NOT in fact encrypt the files, thats why you might find some reports or analysis were the files remained intact. The malware created persistence technique by modifying the registry keys and the logonUI file in addition to placing malicious files in the System32 folder.

8 Appendix

8.1 Appendix A: Volatility Commands Used

```
1 # Image Information
2 vol -f infected-memory.raw windows.info
3
4 # Process Analysis
5 vol -f infected-memory.raw windows.pslist
6 vol -f infected-memory.raw windows.pstree
7 vol -f infected-memory.raw windows.psscan
8
9 # Command Lines
10 vol -f infected-memory.raw windows.cmdline
11
12 # Network Analysis
13 vol -f infected-memory.raw windows.netscan
14
15 # DLL and Module Analysis
16 vol -f infected-memory.raw windows.dlllist
17 vol -f infected-memory.raw windows.modules
18
19 # Registry Analysis
20 vol -f infected-memory.raw windows.registry.hivelist
21 vol -f infected-memory.raw windows.registry.printkey
22
23 # File Extraction
24 vol -f infected-memory.raw windows.procdump
25 vol -f infected-memory.raw windows.dumpfiles
```

8.2 Appendix B: Screenshots

Hash values:

- SHA-1 f368916850332757d7ed2f0ee335c16b9c9fc95b
- SHA-256 d4c83205cf6f3098ab6a757312525f4d14a57a819306eeea5c0d022b00b38cf3
- MD5 1b4f89bdb12a349de92ca7f1261e67a0

Figure 3: Volatility pslist

Figure 4: Volatility pstree

Figure 5: Volatility malfind

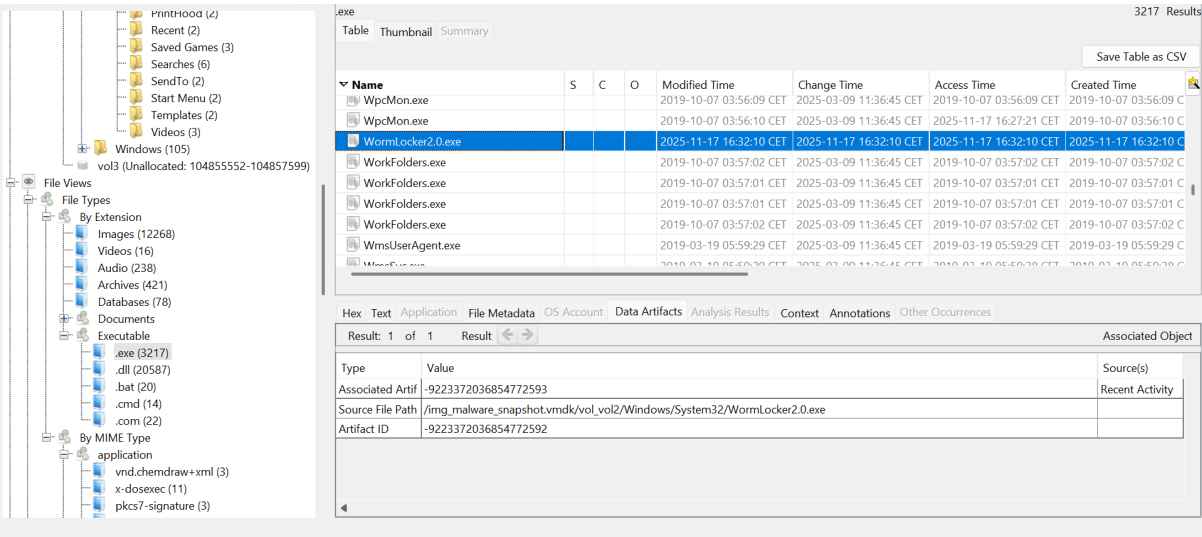


Figure 6: Autopsy file browser

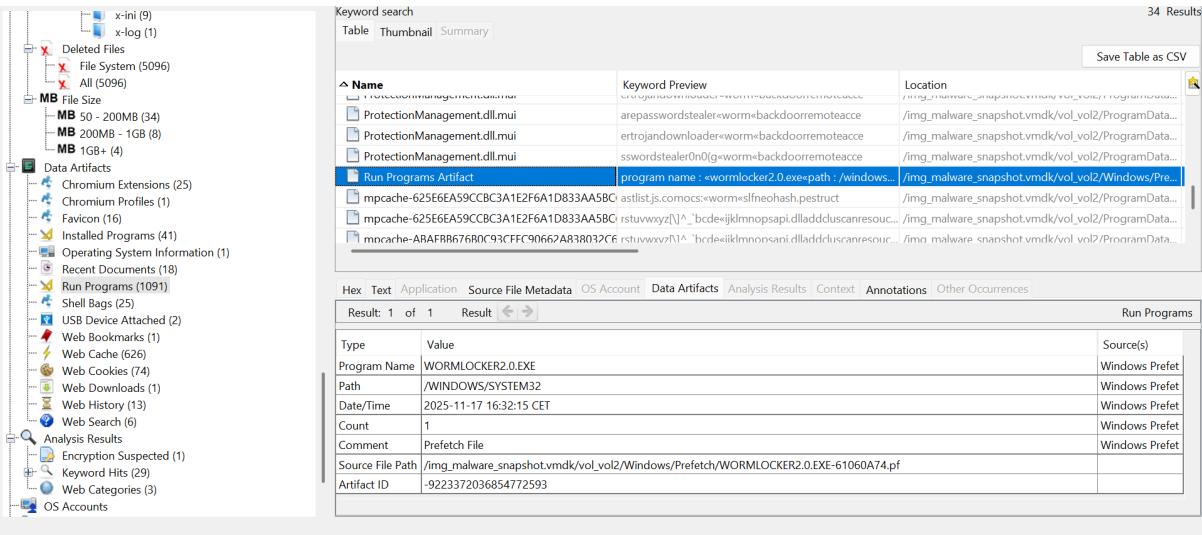


Figure 7: Registry Keys

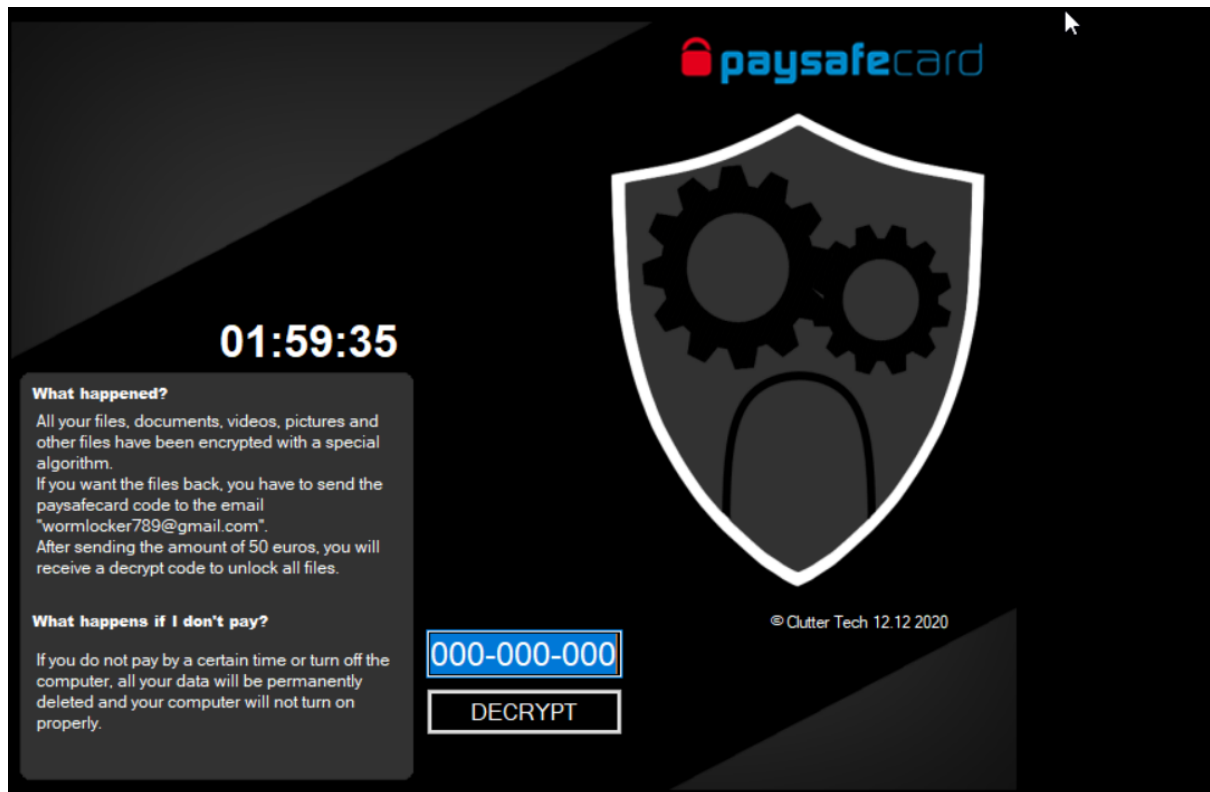


Figure 8: Extracted Ransom Note

9 Ethical and Legal Justification

The malware analysis presented in this project was executed in a fully controlled and isolated environment for educational purpose only. We have considered the following measures:

- **Controlled Environment:** Every execution was made in a virtual machine isolated from the network and from the host (clipboard not shared and no shared folders).
- **Educational Purpose:** This malware simulation was for learning purpose only to understand its behavior and strengthen our defensive capabilities, we do not aim to harm any authority with this.
- **Compliance:** This research complies with institutional policies and applicable laws regarding malware research and cybersecurity education.
- **Responsible Disclosure:** Any vulnerabilities or findings will be handled responsibly and not disclosed publicly without proper authorization.

This analysis follows ethical guidelines for malware research and aims to contribute to the field of cybersecurity defense.

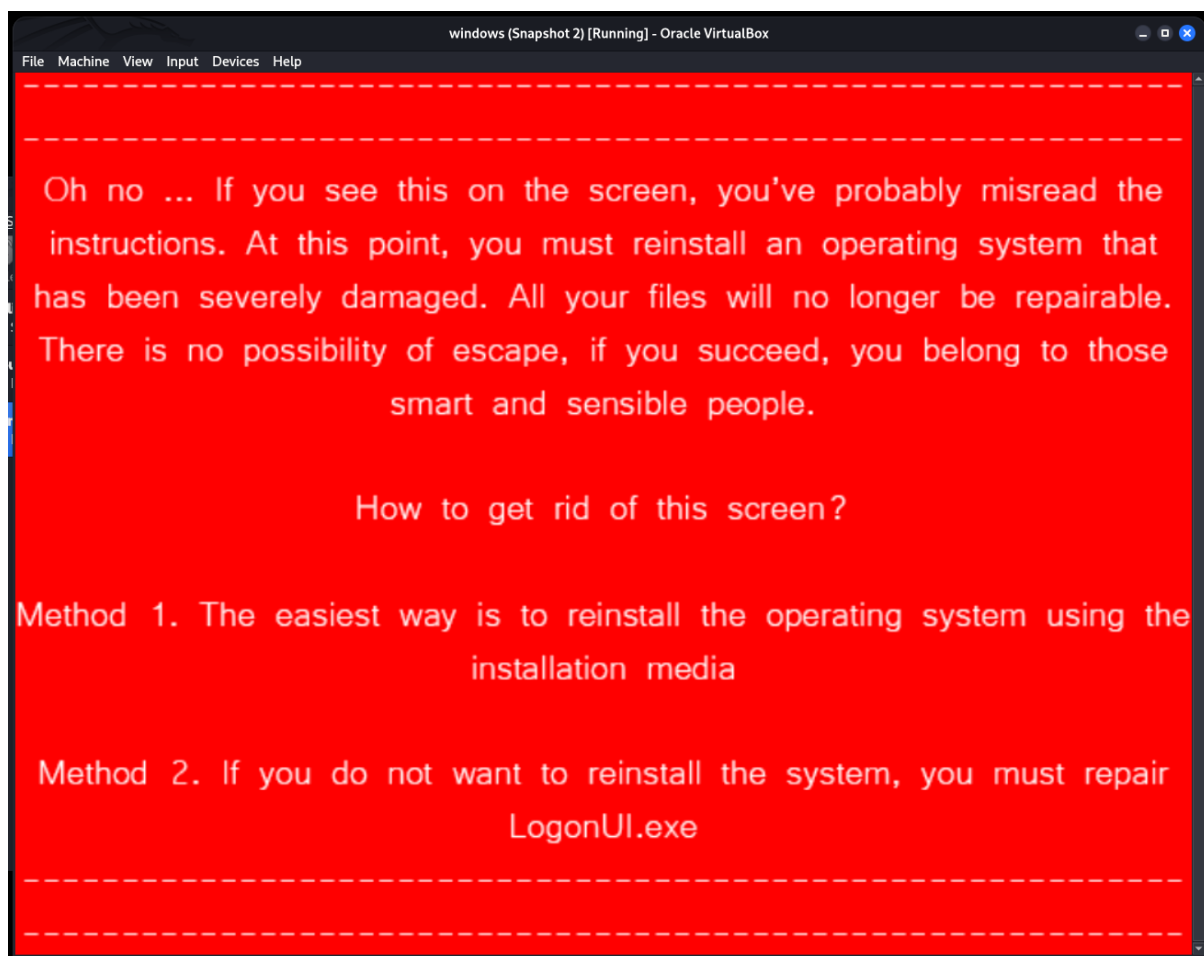


Figure 9: Extracted ransom note after reboot